

WHY GIT COMMIT IS REQUIRED – EASY LEARNING NOTES

1. Why do we need GIT Commit?

GIT Commit is required to save a snapshot (version) of our project. It records changes permanently in GIT history. Without commit, changes are not saved in version control.

2. Important areas in GIT

When we create a project folder, GIT works with three main areas:

Working Directory

This is where we create or modify files. Files here are NOT tracked automatically by GIT.

Staging Area

This is an intermediate area before commit. Files must be moved here using *git add*. (Staging area details are stored inside the .git folder.)

3. How GIT tracks files

New or modified files first stay in the Working Directory. Only files added to the Staging Area are tracked by GIT.

4. git status command

Shows untracked files and modified files. It also guides you to use *git add* when needed.

5. git add command

Moves files from Working Directory to Staging Area, preparing them for commit.

6. git commit command

Saves the staged files permanently using:

git commit -m "Commit message"

7. What happens internally during commit

GIT generates a unique 40-character commit ID (SHA-1). Usually only the first 7 characters are shown for reference.

8. What happens after commit

If the file is modified again, GIT marks it as modified. You must repeat *git add* and *git commit* to save the new version.

9. Simple GIT Flow

Working Directory → git add → Staging Area → git commit → Saved in GIT History

10. Viewing Commit History using git log

The *git log* command is used to view the complete commit history. It shows: • Who committed the changes • When the commit was made • The commit message • The unique commit ID

END OF NOTES